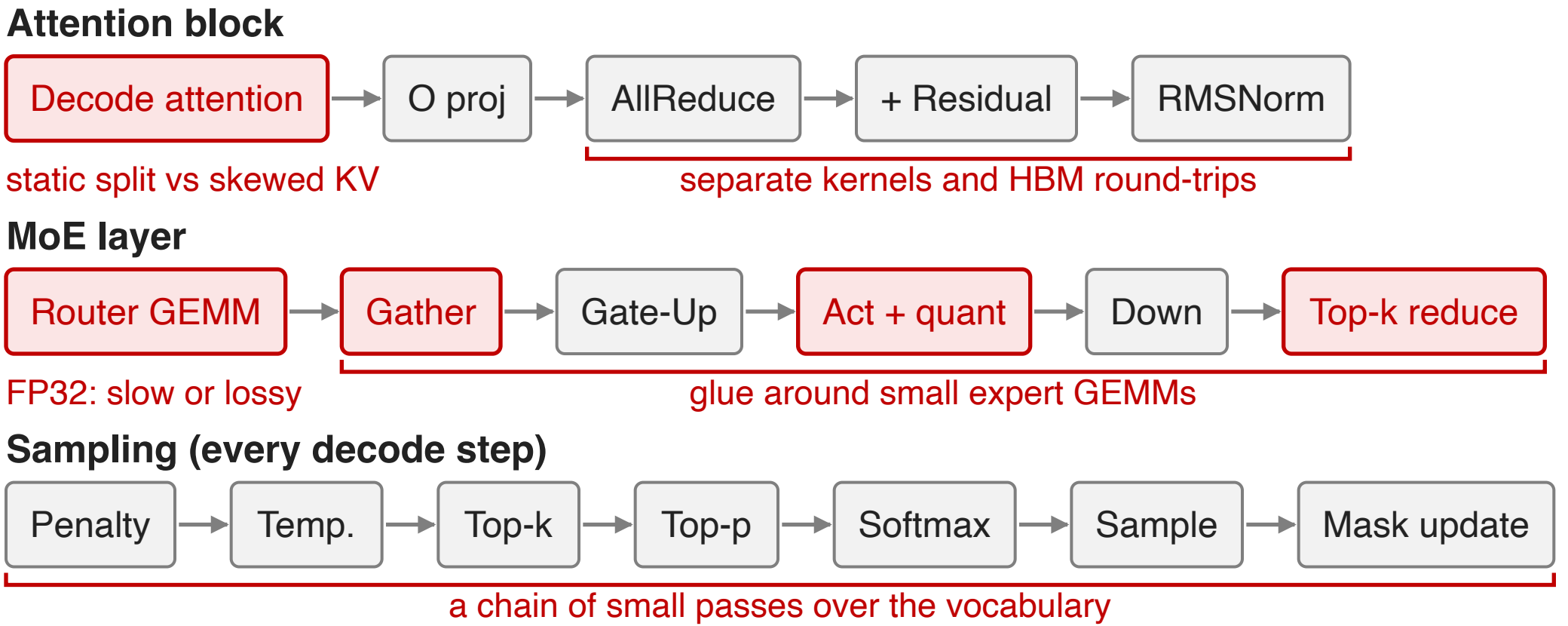
Attention block
Decode attention
O proj
AllReduce
+ Residual
RMSNorm
static split vs skewed KV
separate kernels and HBM round-trips
MoE layer
Router GEMM
Gather
Gate-Up
Act + quant
Down
Top-k reduce
FP32: slow or lossy
glue around small expert GEMMs
Sampling (every decode step)
Penalty
Temp.
Top-k
Top-p
Softmax
Sample
Mask update
a chain of small passes over the vocabulary